

[New Graph Data Model \(draft\)](#)
[New Input Files](#)
[Old Graph Data Model](#)
[Old Input Files](#)
[Data Products](#)
[Notes on Taxonomic Requirements](#)
[Notes on Other Stuff](#)

New Graph Data Model (draft)

Data Sources: traits, TB reference hierarchy, Zenodo record, terms file, References file, autogenerated

Resource

Relationships	Properties
none	resource_id (Zenodo DOI) resource_name

Taxon

Relationships	Properties
parent :Taxon accepted :Taxon trait :Trait inferred_trait :Trait	tbTaxonID scientificName scientificNameID rank higherClassification ???

Trait

Relationships	Properties
predicate :Term supplier : Resource object_term :Term (value) units_term :Term normal_units_term :Term sex_term :Term lifestage_term :Term statistical_method_term :Term source	tb_pk (= TB measurement ID, needed?) resource_pk (= resource measurement ID) scientific_name (= scientificName from trait data source) measurement (value from trait data source for numerical data) object_page_id (for associations) literal (are we still using literals?) normal_measurement (normalized numerical value) normal_units measurementRemarks

reference	
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Term

Relationships	Properties
parent_term:Term synonym_of:Term	uri name name_<locale> section_ids[comma-separated] type (measurement, association, value, metadata) definition, comment attribution position (still needed?) comment exclusive_to_clade incompatible_with_clade object_for_predicate (still needed?)

Scientific Work

Relationships	Properties
	scientificWork_id (DOI, uri, hash) bibliographicCitation

New Input Files

Still mostly based on [DarwinCore](#)

Traits - required elements

dwc:measurementID

dwc:occurrenceID optional

dwc:verbatimTaxonID taxon identifier used by the source, may link to a Taxa file with additional taxonomic data

dwc:taxonID taxon identifier represented in the TB reference hierarchy, imported from Taxa file if present

dwc:verbatimScientificName name used by the source

tb:infer if true, paint the branch: taxa descending from the tbTaxonMapping should inherit this trait

tb:exclude stop branchpainting: descendant taxa should not inherit this trait

dwc:measurementType uri represented in the TB terms file

dwc:measurementValue uri represented in the TB terms file

dwc:measurementRemarks

dwc:measurementUnit uri represented in the TB terms file

dwc:lifestage uri represented in the **TB terms** file (not a label)

dwc:sex uri represented in the **TB terms** file (not a label)

dwc:statisticalMethod uri represented in the TB terms file

dc:source identifier represented in the TB References file

dc:references identifiers represented in the TB References file

Taxa - optional, not harvested into the graph

dwc:taxonID taxon identifier used by the source
tb:taxonID taxon identifier represented in the TB reference hierarchy
dwc:canonicalName derived from the source
dwc:scientificName name used by the source
dwc:parentNameUsageID derived from the source
dwc:higherClassification derived from the source
dwc:kingdom, dwc:phylum, dwc:class, dwc:order, dwc:family, dwc:genus, species derived from the source
dwc:taxonRank derived from the source
dwc:taxonRemarks
dc:source

Occurrences - optional not harvested into the graph

dwc:occurrenceID
dwc:individualCount
dwc:locality
dwc:decimalLatitude
dwc:decimalLongitude
etc.

TraitBank Terms File- one global file for all term mappings (OWL-like)

attribution
definition
name
type
uri
parent uris
synonym of uri
units term
alias
symmetrical assoc
inverse of
incompatible with clade
exclusive to clade

TraitBank Reference Hierarchy - one global file for all taxon mappings

dwc:taxonID
dwc:acceptedNameUsageID
dwc:parentNameUsageID
dwc:scientificName
dwc:canonicalName
dwc:scientificNameAuthorship
dwc:taxonRank
dwc:taxonomicStatus
dc:source
dwc:higherClassification

References - one global file for all sources and references

dc:identifier
dc:bibliographicCitation

Old Graph Data Model

Data Sources: trait data file, EOL managed hierarchy, Zenodo record, autogenerated, terms file

Resource

Relationships	Properties
none	resource_id

Taxon (formerly Page)

Relationships	Properties
parent:Taxon trait:Trait inferred_trait:Trait vernacular:Vernacular (remove)	page_id canonical (remove?) rank

Vernacular (remove)

Relationships	Properties
none	supplier, string (remove) language_code (remove) is_preferred_name (remove)

Trait

Relationships	Properties
predicate:Term supplier:Resource metadata:MetaData object_term:Term (value) units_term:Term normal_units_term:Term sex_term:Term lifestage_term:Term statistical_method_term:Term contributor:Term determined_by:Term compiler:Term	eol_pk (= EOL measurement ID) resource_pk (= resource measurement ID) scientific_name (= scientificName from trait data source) source citation measurement (value from trait data source for numerical data) object_page_id (for associations) literal normal_measurement (normalized numerical value) normal_units method remarks sample_size

MetaData

Relationships	Properties
predicate:Term object_term:Term units_term:Term	eol_pk measurement literal

Term

Relationships	Properties
parent_term:Term synonym_of:Term	uri name name_<locale> section_ids[comma-separated] type (measurement, association, value, metadata) definition, comment attribution is_hidden_from_overview is_hidden_from_glossary position (still needed?) comment exclusive_to_clade incompatible_with_clade object_for_predicate (still needed?)

Old Input Files

required element | not required, imported to database if present | not required not imported to database

Trait Elements

measurementID

parentMeasurementID

measurementType

measurementValue

measurementUnit

Metadata

measurementAccuracy

measurementMethod

measurementRemarks

statisticalMethod

sex

lifeStage

location

Associations

associationID

associationType

Taxonomic Elements

taxonID [eolID, link to the EOL hierarchy]

scientificName [from trait data source]

other taxonomic data provided by the trait data source, e.g., rank, higherClassification, taxonomicStatus

Attribution Data

source

bibliographicCitation

references

contributor

compiler

determinedBy

References

referenceID

fullReference

New TraitBank Input Files

- A resource archive might contain only the M&F file = Trait file.
 - Optional additional files include Taxon and Occurrence.
- Only the Trait file is harvested into the graph
 - Data from the Taxon & Occurrence files will not be accessible through graph queries and will not be included in query results. But they will be available in the resource file on Zenodo.
 - We assume that these data will not be of interest to most data users.
 - A property on the trait data record should alert data users about the availability of additional Taxon or Occurrence information in the resource file on Zenodo.
 - This will make this information available to the few data users who may want to drill deeper into the source data.
 - In the graph, this would be a resource level property that propagates to traits

Trait File

- The Trait file contains all data destined for the graph and possibly additional columns for a human downloader to read.
- Trait files may have a *taxonID* column
 - If the source is a database that provides taxon identifiers, we should use these identifiers

- as the *taxonID* value
 - For most resources, the *taxonID* will be the scientific name string used to identify the taxon in the source work. The name string can include author information or it can be a canonical. Whatever is provided by the source.
 - The main function of the *taxonID* column is to link the trait record to a record in the *Taxon* file
 - If there is no taxon file associated with a resource, we don't need the *taxonID* column.
 - Should we harvest *taxonID* values into the graph? They would be useful if they are real identifiers. If they are just name strings, they are redundant with *scientificName*.
- Trait files will have a *scientificName* column. This preserves the name attached to the trait record in the source work.
 - This may be the canonical name or the full scientific name with author data, whatever is easier to grab/mine from the resource.
 - *scientificName* values are harvested into the graph and included in trait data products.
- Trait files will have an *tbTaxonID* column that links to *taxonID* values in the TraitBank management hierarchy whenever possible.
 - *tbTaxonID* values that are simple integers link to the management hierarchy. Records with these values are harvested into the graph.
 - *tbTaxonID* values that are not simple integers (i.e., that are blank or have other strings) represent unmapped nodes. Records with these values are not harvested into the graph.
 - Records for unmapped nodes are in the *Trait* file in order to represent all trait information from the source.
 - Unmapped nodes may become mappable with additional research.
- Trait files may have an *occurrenceID* column if there is additional information in an external Occurrences file, but commonly used occurrence columns, like life stage or sex can also be included in the Trait file.
- Trait files have a source column with the identifier (doi, url, handle) of the source work. Bibliographic citations will be maintained in a References masterfile. Most will be imported from CrossRef or WikiData, so we won't need to add those in the resource file anymore.

Taxon File

- If the source provides parent child relationships, authority info or anything else that may be considered of possible interest to data users, these data can be provided in the taxon file for a human downloader, but this information will not be represented in the graph.
 - Not all *Trait* files need a *Taxon* file
 - If there is a *Taxon* file, *eoIID* values in the *Taxon* file take priority over *eoIID* values in the *Trait* file, i.e., we need a script to automatically load *eoIID* from the *Taxon* file into the *Trait* file.
- Data in the Taxon file should reflect names and relationships as represented in the source work.
 - The *taxonID* links to the *Trait* file.
 - The *eoIID* links to the management hierarchy
- The Taxon file is not harvested into the graph.

Occurrence File

- If the source provides multiple redundant trait data ([see Occurrence Data](#)), they can be included in detail here. A property on the trait record will point to this.

References

- In the new graph, we should have just one kind of object for sources & references.
 - What should we call this object? <https://www.wikidata.org/entity/Q11826511> (scientific work)
- This object will have two properties:
 - identifier, i.e., doi, handle, wikidata uri for things that don't already have an identifier
 - citation for the bibliographic citation - > imported from CrossRef or WikiData
- Scientific Work objects will be linked to trait data records through two alternative relationships:
 - <http://purl.org/dc/elements/1.1/source>
 - <http://purl.org/dc/terms/references>
- All Scientific Work objects will be managed in one master References file with two columns: identifier, citation
 - When the data are harvested, the validation step for trait files should include a step where it checks if all source/reference IDs are covered in the References master file. If not, they should be added, and ideally, the citation should be resolved automatically from CrossRef, WikiData and other resources.
 - Some identifiers may not be able to be resolved to a citation and will have to be added manually by Jen or Katja. For this, we should have a simple interface, where we get a list of rows in the References master file where the citation is empty, along with a field where we can add the missing citations and send them to be added to the file.
 - For references that don't have any identifiers, we will make up globally unique identifiers, e.g., by using a hash.
 - These identifiers along with their citation values will be stashed in a document somewhere.
 - The validation process for the references will check this document to fill in missing citations.
 - The interfaces for missing citations and manually generated identifiers can probably be combined in a single document where the validation process picks up new references and deletes the ones that it has processed and drops off identifiers in need of a manually created citation.

Going forward:

- Use DOIs whenever possible for source & referenceID.
 - Reference validation process will resolve DOIs and add citations to the References master file.
- If no globally unique identifier is available:
 - create a WikiData record for the reference, which will be resolved by reference validation process OR

- create an identifier (i.e., hash) and add the reference & citation to the references-without-identifier list.
- If there is another globally unique identifier like a handle available, use that identifier. If it doesn't resolve to a citation, add the citation once it shows up in the references-without-citation list.

Data Products

taxonID [eolID = pageID]

scientificNameID [link to the external reference hierarchy]

higherClassification [according to the EOL hierarchy]

scientificName [according to the trait data source]

scientificName [according to the EOL hierarchy]

measurementType

measurementTypeURI

measurementValue

measurementValueURI

measurementUnit

measurementUnitURI

measurementRemarks

measurementMethod

statisticalMethod

targetTaxon

targetTaxonSourceName

targetTaxonName

sex

lifeStage

source

bibliographicCitation

referenceID

resourceURL

Notes on Taxonomic Requirements

Accepted Names in the Graph

We currently have two kinds of names in the graph: the preferred canonicalName for the mapped eolID and the scientificName represented in the trait data file. I think it may be a good idea to use the EOL scientificName rather than the canonical in the graph. This would make it easier to distinguish homonyms, but it may make querying more complicated because people usually search using canonical strings, not full scientificNames. Maybe we could have both in the graph, if that doesn't create too much overhead? Consult with Jorrit.

Synonyms in the Graph

Do we want to have synonyms in the graph? They could be useful if we want to let people search on them.

If we want synonyms, how should we treat them in the data model? Treating them as properties would be simpler, but then we couldn't have their scientificNameIDs and other metadata in the graph. Is it enough if this information is available in the EOL hierarchy file?

Current thoughts: We should treat synonyms as nodes in the graph, and we should allow traits to be attached to synonyms in the Traits file. In the graph, the trait data would then be attached to the acceptedName via its synonym. This would automatically transfer the trait mapping to the new acceptedName when the hierarchy is updated. We currently have many traits where the source uses a synonym, and we have manually mapped it to the current accepted name. We could gradually clean those up, but there will probably always be some source names that we will not want to map as synonyms, e.g., gross misspellings.

HigherClassification Data in the Graph

Should we store higherClassification ID strings in addition to parent taxa in the graph? This would make the graph a lot larger, but it may make queries more efficient. Consult with Jorrit.

Legacy Taxa

There will be taxa in our trait data sets that cannot be cleanly mapped to any accepted taxa, because there are no synonym mappings available and/or mapping would require a deep understanding of the taxon concept used in the trait data source.

For species it's probably ok to tentatively place the name in the EOL hierarchy, even if we know it's not in current use. This preserves the trait data record, and we can always change the mapping later if we can figure it out.

But this is a real problem for inferred traits from older sources, e.g., the genera from the Kubitzki data set. How should we handle genera and other higher taxa if we know the trait data set name is not in use anymore or its concept has changed significantly? I don't think we should account for this situation in the data model, but we should talk about best practices. For example, we may decide to include data for such taxa as individual data points, but not use them for taxonomic inference. Data like this may be useful for data users who have the knowledge to do their own mappings of taxon concepts.

Duplicate Taxa in Catalogue of Life

There's a lot of taxon duplication in Catalogue of Life. There are things like unrecognized synonyms, alternative representations from WoRMS, whole branches imported twice from different suppliers with different names at the root but overlapping genera and species. This often happens in branches with fossil taxa. I am inclined to omit these taxa from the EOL hierarchy, but if I do this, I will have to pick a winner, and I need to develop a procedure to mark the losers, so we know they have been excluded as duplicates in future updates. There are also complications with merging the descendants of partially duplicated branches. If I move some taxa from one COL ancestor to another, I am changing the reference hierarchy, which takes us back into more serious hierarchy management territory. I was trying to avoid this, but it may be necessary.

Taxonomic Data from Trait Data Sources

Taxonomic data provided by the trait data source could go into a separate Taxon file. I don't think these data should go into the graph, but they could be harvested from the resource file for extended data products.

Note that there may be scenarios where the taxonomic data provided by the trait data source conflicts with the EOL hierarchy. This will not be apparent in the resource file and may lead to confusion.

Taxon Documentation in Data Products

For names, we want both the EOL `scientificName` and the `scientificName` from the trait data source. Higher classification (according to our management hierarchy) could be an optional field. What do we want to provide in terms of identifiers? In addition to the TraitBank `taxonID` (=eolID), we may want to provide the `scientificNameID` which is the link to the external hierarchy resource (usually COL).

What if the trait data file has GUIDs as `taxonIDs` or provides their own mapping to external GUIDs outside EOL. Shouldn't we include those in data products? If so, we either need to add them to the trait properties.

Transparency for Data Users

We should include an `eml` file or something similar in both the resource files and the data packages that explain our taxonomic mapping rationale along with other things like the ontology mappings.

Taxonomic Data in Input Files

Initial scenario, now dismissed:

- Trait files will have a *taxonID* (=eolID) column that links to *taxonID* values in the management hierarchy whenever possible.
 - *taxonID* values that are simple integers link to the management hierarchy
 - *taxonID* values that are not simple integers represent unmapped nodes
 - These are in the Trait file in order to represent all trait information from the source, but they will not go into the graph
 - Unmapped nodes may become mappable with additional research
 - The *taxonID* values of unmapped nodes can be used to link to additional taxonomic information in the Taxon file.
- Trait files will have a *scientificName* column. This preserves the name attached to the trait record in the source work.
 - This may be the canonical name or the full scientific name with author data, whatever is easier to grab/mine from the resource.
- Taxon file:
 - The taxonID links to the management hierarchy
 - Mapping nodes from the management hierarchy can be included to support complex taxon mappings for synonyms and other names not represented in the management hierarchy.
 - What if the source work provides GUIDs as taxonIDs or provides their own mapping to external GUIDs outside EOL? This will be rare.
 - We could represent this information with additional identifier columns.
 - Should we have this information in the graph, i.e., included in standard data products?

Notes on Other Stuff

Geographic Data

- Going forward, only harvest geographic data that is expert curated, eg., from [curated databases](#), published country checklists, etc.
- Make sure charismatic groups like mammals & birds are well covered.
- Then do a gap analysis to see how well we are doing.

“Occurrence” Data

- Do we want occurrence metadata in the graph?
 - Probably not. That would add a lot of extra columns that are only used rarely.
 - We also don't want to have a lot of redundant individual measurements for a given taxon.
 - Graph data should generally be summary data for the species, broken down by sex, life stage, geographic region if there are differences
 - Rather than weighing the graph down with lots of occurrence level metadata, we could

keep these data in the resource file and link to it in the graph output.

- occurrences are “individual measurement” data records which usually have occurrence level metadata
- If there are multiple individual measurements for a species with similar metadata (lifestage, sex, location) in a given data file, we want to have a summarized version of these data in the measurements & facts file, while the individual measurements and their detailed metadata stay in the occurrences file.
- The occurrences file is included in the main resource archive, and is available redundantly as a separate file in the same zenodo dataset.
- The occurrences file includes a column called “summarized by” or something, containing the measurement ID of the summary record in the M&F file.
- The information from the occurrences file does not go into the graph, but the record with the summarized version will have a property, populated from a column in the M&F file, called “additional metadata” or something, with a term definition explaining the relationship, and a value of [URL pointing to the occurrences file].

List of tasks for Eli’s validation script

When a trait data file is ready for publication, we should run a validation script on it to make sure there aren’t any loose ends:

1. If there is a Taxon file, update the eoIID values in the Trait file to match the ones in the Taxon file.
2. Trait file validation:
 - a. eoIID values should match taxonID values in the reference hierarchy -> report values that don’t have a match.
 - b. stop nodes should descend from start nodes -> report start/stop node pairs that are in violation.
 - c. uris in the trait, value, lifestage, sex, columns should match uris in the terms file -> report uris that don’t have a match
 - d. referenceIDs should have an identifier match in the References master file -> automatically add missing identifiers, try to fetch bibliographic citation via CrossRef, WikiData, or another service, add identifiers that cannot be resolved to a manual reference curation doc.
- 3.